Francisco Pereira

✉ francisco.chamica.pereira@tecnico.ulisboa.pt github.com/fchamicapereira https://fchamicapereira.me
Office 601, INESC-ID, Rua Alves Redol 9, 1000-029, Lisbon, Portugal
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Education

Instituto Superior Técnico (University of Lisbon) 2021 - Present
Ph.D. in Computer Science and Engineering
Carnegie Mellon Portugal Affiliate Program
Advisors: Prof. Luis Pedrosa, Prof. Fernando Ramos, Prof. Justine Sherry

Instituto Superior Técnico (University of Lisbon) 2019 - 2021
M.Sc. in Computer Science and Engineering
Advisors: Prof. Luis Pedrosa, Prof. Fernando Ramos
Thesis: Parallel Network Function Synthesis

Instituto Superior Técnico (University of Lisbon) 2016 - 2019
B.Sc. in Computer Science and Engineering

Research Experience

Research Assistant, *Carnegie Mellon University, Pittsburgh, PA, USA* Feb 2024 - Feb 2025

Research Intern, *Telefonica, Barcelona, Spain* Oct 2023 - Jan 2024
Designing a network traffic change detector completely on the dataplane capable of withstanding Tbps traffic. More specifically, an implementation of the k-ary algorithm for the Tofino programmable switch architecture.

Research Assistant Intern, *Carnegie Mellon University, Pittsburgh, PA, USA* Jul 2022 - Oct 2022

Research Interests

Networking and Computer Systems. Program analysis and automatic manipulation of software network functions with strong enphasis on performance. Automatic offloading of NFs to hardware network accelerators (FPGAs, SmartNICs, and Programmable Switches). Hardware design for network function acceleration.

Scholarships and Awards

- CMU Portugal Affiliate Ph.D. Program, FCT (Fundação para as Ciências e Tecnologias) 2021 - Present
- Master Student Merit Award 2020, Distributed Systems Group @ INESC-ID 2020

Publications

Automatic Parallelization of Software Network Functions 2024
Francisco Pereira, Fernando Ramos, Luis Pedrosa.
21st USENIX Symposium on Networked Systems Design and Implementation (**NSDI'24**).

Automatic Generation of Network Function Accelerators Using Component-Based Synthesis 2022
Francisco Pereira, Gonçalo Matos, Hugo Sadok, Daehyeok Kim, Ruben Martins, Justine Sherry, Fernando Ramos, Luis Pedrosa.
ACM SIGCOMM Symposium on SDN Research (**SOSR'22**).

Selected Posters

In-Network ML Feature Computation for Malicious Traffic Detection 2023
João Amado, **Francisco Pereira**, Salvatore Signorello, Miguel Correia, Fernando Ramos.
ACM SIGCOMM'23 Posters and Demos.

Projects

SyNAPSE, DPSS @ INESC-ID & SNAP Lab @ CMU 2021 - Present
When implementing network functions (NFs), developers are often confronted with a choice: implement the NF in software and face the challenge of performance, or use one of a wide variety of programmable networking devices, such as programmable switches and SmartNICs, to trade-off some flexibility for the ability to process packets at full line rate. With SyNAPSE we ask "Why not have both?". We propose a synthesis based approach to automatically generate accelerated implementations of a Software NFs, using smart network devices whenever possible to increase performance.

Maestro, DPSS @ INESC-ID 2020 - 2023
Software network functions (NFs) trade-off flexibility and ease of deployment for an increased challenge of performance. Fortunately, a number of tools, techniques, and architectural enhancements have come together over the years to enable packet processing at 100s Gbps, but achieving these traffic rates requires a deep understanding of low-level details. The traditional way to increase NF performance is by distributing traffic to multiple CPU cores, but this poses a significant challenge: *how to parallelize an NF without breaking its semantics?* We propose Maestro, a tool that analyzes a sequential implementation of an NF and automatically generates an enhanced parallel version.

Research Talks

Maestro
- USENIX NSDI24 Apr 2024
- MIT CSAIL Apr 2024

Teaching Experience

Distributed Systems Spring 2022
Instituto Superior Técnico (University of Lisbon)
Undergraduate Teaching Assistance

Computer Networks Fall 2022
Instituto Superior Técnico (University of Lisbon)
Undergraduate Teaching Assistance

Computer Networks Fall 2021
Instituto Superior Técnico (University of Lisbon)
Undergraduate Teaching Assistance

Community Service

Organization Committee, Netdev 0x16 2022
HotCRP Chair, EuroP4'21 2021
Network admin, DPSS @ INESC-ID 2021 - Present
Student Volunteer, SIGCOMM'21 2021
DevTeam Team Leader, SINFO 2018 - 2020

Programming Skills

Programming languages

Daily use: C, C++, Python, P4
Comfortable: Javascript, Java
Used in the past: C#, Go
Currently learning: SystemVerilog

Frameworks

DPDK, KLEE, LLVM, Z3, node.js, AngularJS